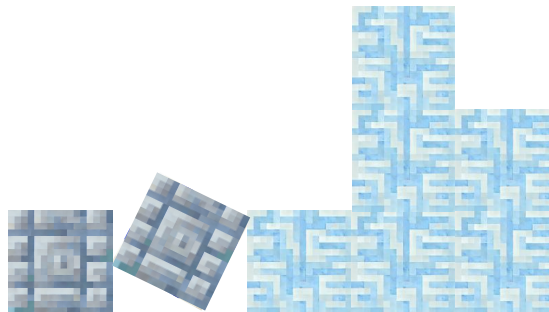


End-to-end ML system for Minecraft texture generation



What is MineTexture?

- A tool to create new textures for **Minecraft**
 - To modify Minecraft default texture
 - To create new items, blocks, etc...

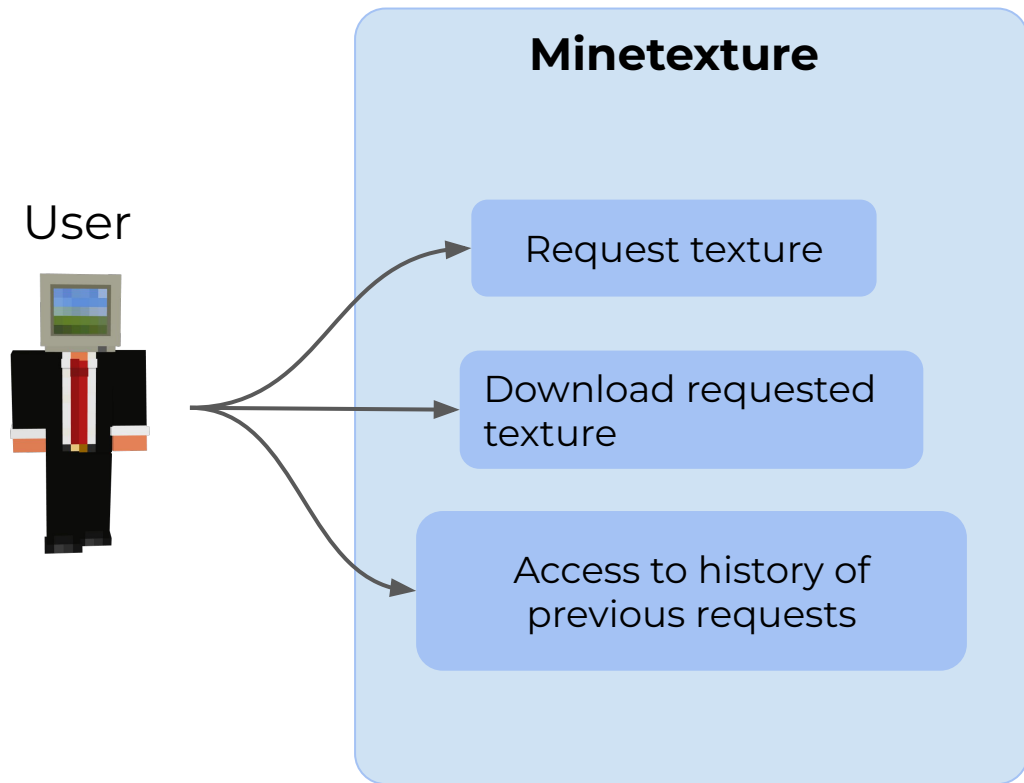


Minecraft with default texture pack



Custom texture packs

Use case

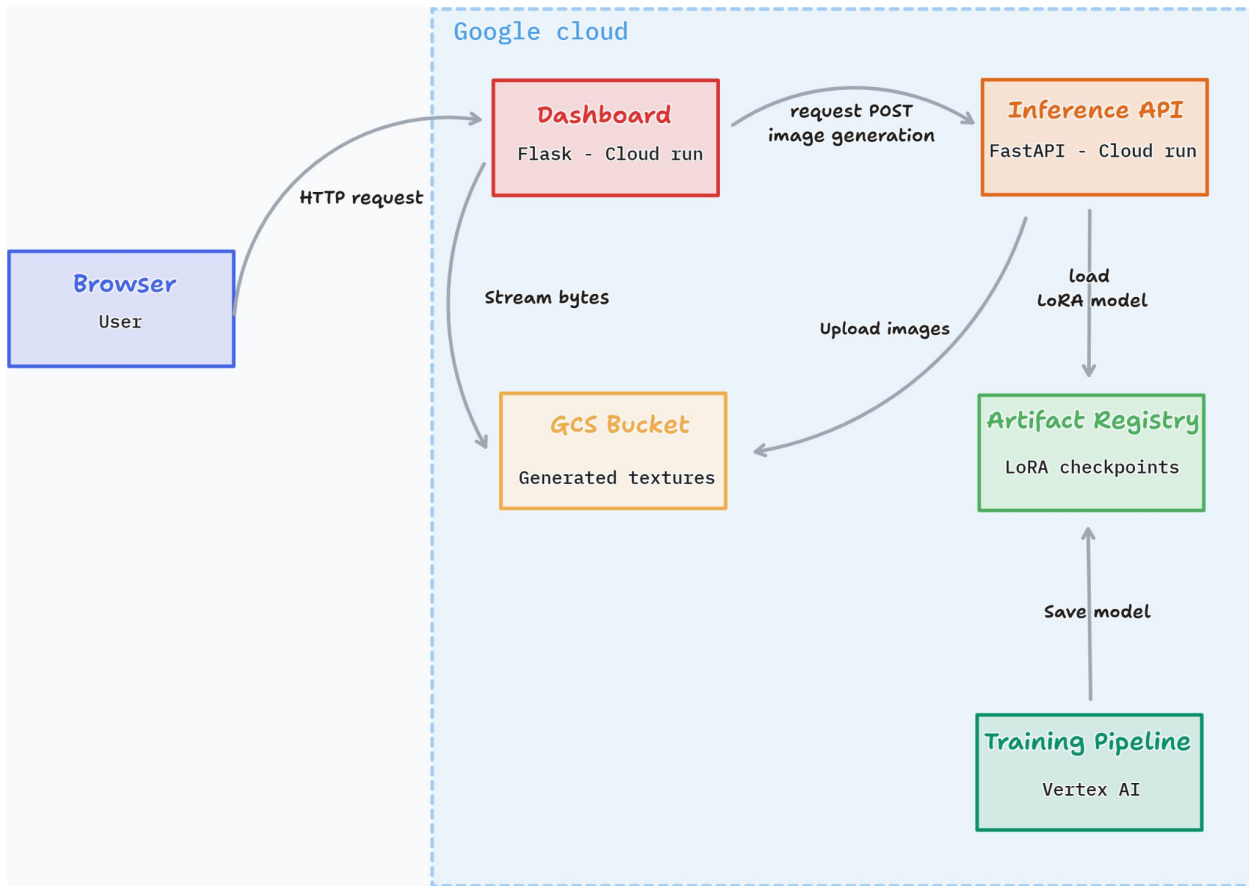


For:

- **Developers** that need quick textures for their mods
- **Texture pack makers** to get ideas for new textures
- **Players** that wants to customize their texture pack

It is **NOT** a replacement for real artists, instead helps on fast iterations

Overall architecture



Training Data: Dataset & preprocessing

Our dataset:

Style learning 🎨

- Texture packs
- Redesign existing objects
- Provide stylistic diversity

Content learning 🧱

- Mods
- Add new objects
- Expand the object vocabulary

→ 30 083 texture images in total

Preprocessing:

Resize

*Nearest
Neighbor*



Bicubic



Background normalization



Final structure



{ashen}-{item}_{melon_slice}.png



{ashen}-{item}_{melon_slice}.txt

Training - Overview

- Qualitative experiments led to choose stable-diffusion 1.5
 - Relatively fast
 - Well documented *style learning* fine tuning pipelines
 - Tested fined tuned models performed as desired
- Standard style learning fine tuning is based on LoRA.
- Slow and long training run
 - Too costly to run on Google Cloud (>600\$)
 - Training on on-premise server



Training - On-Premise - VertexAI



Inference API

Base image: nvidia/cuda:12.1.0-runtime

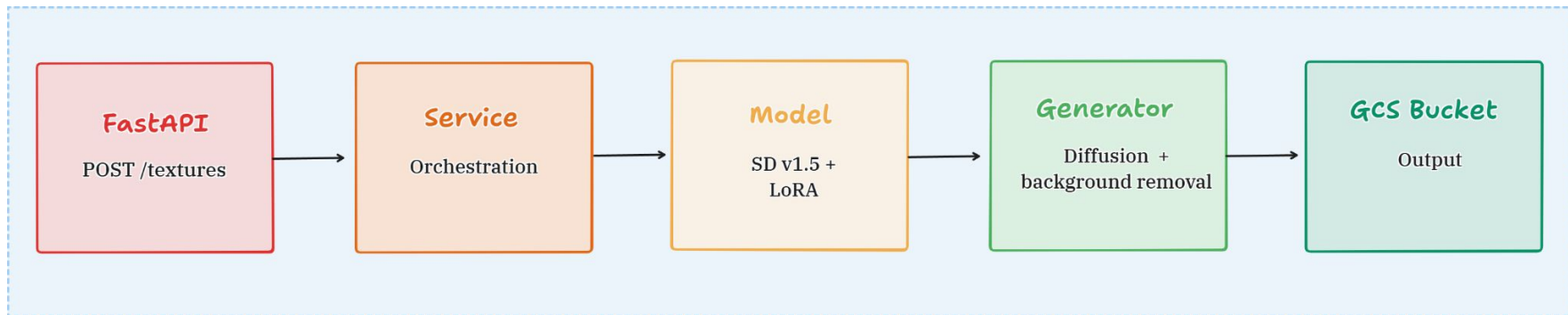
Server: Uvicorn (1 worker)

Compute: GPU required (nvidia-l4)

Scale to 0

Latest check point selection

- {model_name}_{YYYYMMDD}-{HHMM}
- minetexture-lora_20260510-1125



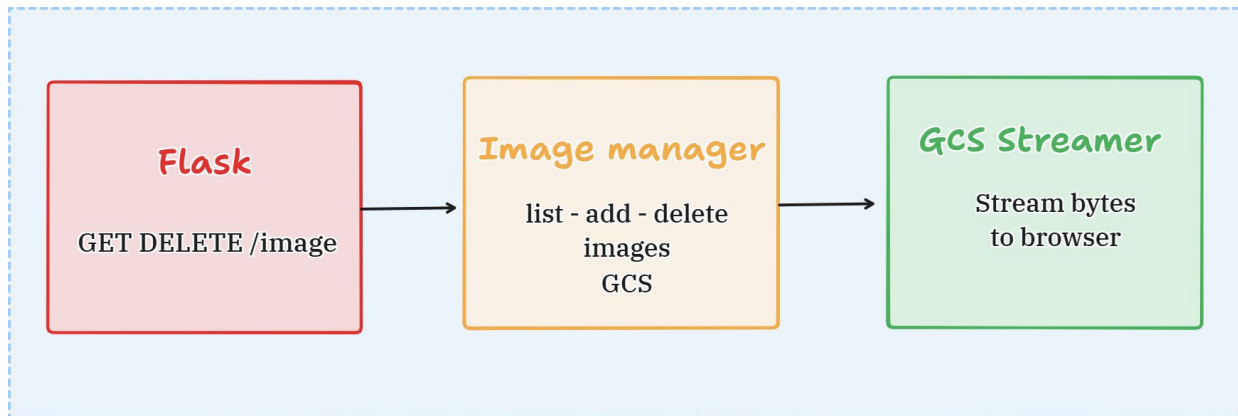
Dashboard

Base image: python:3.11-slim

CPU: 1

Scale to 0

```
gs://generated_data_minetexture/generation/  
generated_data_minetexture/  
└─ generation/  
    └─ <session_id>/  
        └─ <slug>-<timestamp>.png  
            ...
```



Demo



<https://dashboard-217219593769.europe-west1.run.app/>

“Pickaxe item purple
fantasy-tech
magical_fantasy”





Questions?

INFO9023 - Presentation

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